Queueing Systems: Analysis and Simulation

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1 Introduction

Queueing theory studies stochastic systems involving waiting lines and plays an important role in computer systems, telecommunications, and service operations.

Queueing models are commonly described using Kendall notation $A/B/c$, where A denotes the arrival process, B the service time distribution, and c the number of servers. In particular, M denotes exponential (Markovian) interarrival or service times, while G denotes a general distribution.

In this project, we develop the mathematical framework of queueing systems using continuous-time Markov chains (CTMCs), derive key analytical results, and validate them through simulation.

1.1 The Balance Method

The balance method is used to determine the stationary distribution of a stochastic process by equating the flow of probability into each state with the flow out of that state.

In steady state, the probabilities $\{p_j\}$ satisfy

$$(\text{rate of flow into state } j) = (\text{rate of flow out of state } j).$$

For birth–death processes, this leads to the balance equations

$$\lambda_j p_j = \mu_{j+1} p_{j+1},$$

which provide a recursive relationship between successive probabilities.

Solving these equations together with the normalization condition

$$\sum_{j=0}^{\infty} p_j = 1$$

yields the stationary distribution.

2 M/M/1 Queue: Analysis

2.1 Model Description

We consider an $M/M/1$ queueing system.

- Customers arrive according to a Poisson process with rate λ
- Service times are exponentially distributed with rate μ
- There is a single server operating on a first-come, first-served basis

Let $\{X_t\}_{t\geq 0}$ denote the number of customers in the system at time t. The process $\{X_t\}$ takes values in $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.

2.2 Stationary Distribution

Let

$$p_j = \lim_{t\to\infty} \mathbb{P}(X_t = j)$$

denote the long-run probability of being in state j, assuming the limit exists.

Balance Equations

In steady state, the flow into each state must equal the flow out of that state.

For $j \geq 1$, transitions into state j occur from:

- state $j-1$ via arrivals at rate λ
- state $j+1$ via service completions at rate μ

Transitions out of state j occur at total rate $\lambda + \mu$.

Thus, the balance equations are

$$\lambda p_{j-1} + \mu p_{j+1} = (\lambda + \mu) p_j, \quad j \geq 1.$$

For $j = 0$, we have

$$\mu p_1 = \lambda p_0.$$

Solution

A simpler set of equations is given by the detailed balance condition:

$$\lambda p_{j-1} = \mu p_j, \quad j \geq 1.$$

This gives the recursion

$$p_j = \frac{\lambda}{\mu} p_{j-1}.$$

Iterating,

$$p_j = \left(\frac{\lambda}{\mu}\right)^j p_0.$$

Normalization

Using $\sum_{j=0}^{\infty} p_j = 1$,

$$\sum_{j=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^j p_0 = 1.$$

This is a geometric series, which converges if

$$\rho = \frac{\lambda}{\mu} < 1.$$

Thus,

$$p_0 = 1 - \rho,$$

and hence

$$p_j = (1-\rho)\rho^j, \quad j \geq 0.$$

2.3 Expected Number of Customers

The expected number of customers in steady state is

$$\mathbb{E}[N] = \sum_{j=0}^{\infty} j p_j.$$

Substituting $p_j = (1-\rho)\rho^j$,

$$\mathbb{E}[N] = (1-\rho) \sum_{j=0}^{\infty} j\rho^j.$$

Using

$$\sum_{j=0}^{\infty} j x^j = \frac{x}{(1-x)^2}, \quad |x| < 1,$$

we obtain

$$\mathbb{E}[N] = (1-\rho)\frac{\rho}{(1-\rho)^2}.$$

Thus,

$$\mathbb{E}[N] = \frac{\rho}{1-\rho}.$$

Substituting $\rho = \frac{\lambda}{\mu}$,

$$\mathbb{E}[N] = \frac{\lambda}{\mu - \lambda}, \quad \lambda < \mu.$$

2.4 Interpretation

The condition $\rho = \frac{\lambda}{\mu} < 1$ ensures stability of the system. As λ approaches μ, the expected number of customers grows rapidly, indicating increasing congestion in the system.

2.5 Simulation Methodology

We simulate the CTMC and estimate:

- Limiting probabilities p_j using time proportions
- $\mathbb{E}[N]$ using time averages

2.6 R Implementation

We simulate the $M/M/1$ queue as a continuous-time Markov process and estimate key quantities using time averages.

```
# Simulation
simulate_mm1 <- function(lambda, mu, max_events = 100000){
  t <- 0; n <- 0
  times <- numeric(max_events)
  states <- numeric(max_events)

  for(i in 1:max_events){
    rate <- if(n == 0) lambda else (lambda + mu)
    dt <- rexp(1, rate)
    t <- t + dt

    times[i] <- t
    states[i] <- n

    if(n == 0){
      n <- 1
    } else {
      if(runif(1) < lambda/(lambda+mu)) n <- n+1 else n <- n
          -1
    }
  }
  list(times=times, states=states)
}

# Estimation
estimate_pj <- function(times, states, max_state=10){
  durations <- diff(c(0, times))
  total_time <- sum(durations)
  sapply(0:max_state, function(j)
    sum(durations[states==j]) / total_time)
}

estimate_EN <- function(times, states){
  durations <- diff(c(0, times))
  sum(states * durations) / sum(durations)
}
```

2.7 Results

```
lambda <- 2; mu <- 3

sim <- simulate_mm1(lambda, mu)

p_hat <- estimate_pj(sim$times, sim$states)
EN_sim <- estimate_EN(sim$times, sim$states)
```

The simulated estimates are compared with the theoretical values

$$p_j = (1-\rho)\rho^j, \qquad \mathbb{E}[N] = \frac{\lambda}{\mu - \lambda}.$$

The estimated and theoretical values of the expected number of customers are:

$$\text{Simulated } \mathbb{E}[N] \approx 1.9627, \qquad \text{Theoretical } \mathbb{E}[N] = 2.$$

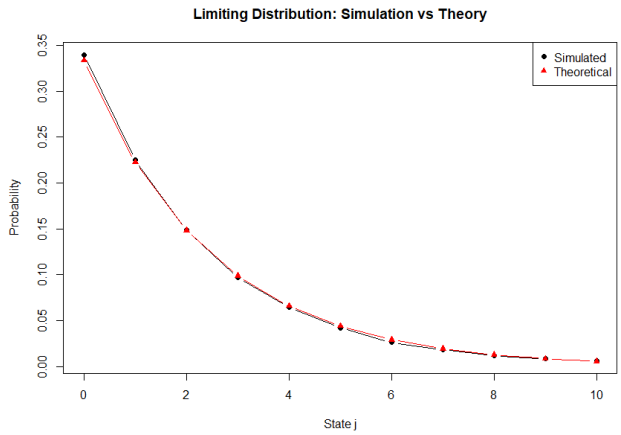
The results show close agreement between simulation and theory. The results show close agreement between simulation and theory.

2.8 Visualization

```
# Plot comparison
plot(0:max_state, p_hat, type="b", pch=16,
     xlab="State j", ylab="Probability",
     main="M/M/1: Simulation vs Theory")

lines(0:max_state, p_true, col="red", type="b", pch=17)

legend("topright",
       legend=c("Simulated", "Theoretical"),
       col=c("black", "red"),
       pch=c(16,17))
```



2.9 Conclusion

The simulation results closely match the theoretical expressions

$$p_j = (1-\rho)\rho^j, \qquad \mathbb{E}[N] = \frac{\lambda}{\mu - \lambda}.$$

The empirical distribution of states obtained from the simulation aligns well with the geometric stationary distribution, confirming that the system converges to its steady-state behaviour over time.

Furthermore, the simulated estimate of $\mathbb{E}[N]$ is in strong agreement with the theoretical value, validating the time-average approach used in the simulation.

These results support the assumption that

$$p_j = \lim_{t\to\infty} \mathbb{P}(X_t = j)$$

exists and is independent of the initial state. This demonstrates the ergodic nature of the $M/M/1$ queue under the stability condition $\rho < 1$.

Finally, we observe that as λ approaches μ, the expected number of customers increases significantly, highlighting the impact of system load on congestion.

3 M/M/c Queue: Analysis

3.1 Model Description

We now consider an $M/M/c$ queueing system with c identical servers.

- Arrivals follow a Poisson process with rate λ
- Service times are exponentially distributed with rate μ

- There are c parallel servers

Let $\{X_t\}_{t\geq 0}$ denote the number of customers in the system.

3.2 Transition Structure

When the system is in state i:

- Arrivals occur at rate λ, so

$$P_{i,i+1}(h) = \lambda h + o(h)$$

- Departures occur at rate $\min(i,c)\mu$, since at most c customers can be served simultaneously.

  When the system is in state i, there are $\min(i,c)$ customers in service. Each customer in service has an exponentially distributed service time with rate μ, and these service times are independent.

  Let $S_1, S_2, \ldots, S_{\min(i,c)}$ denote the remaining service times. The time until the next departure is given by

$$\min\{S_1, S_2, \ldots, S_{\min(i,c)}\}.$$

  Since the minimum of independent exponential random variables with rate μ is again exponential with rate equal to the sum of the rates, we have

$$\min\{S_1, \ldots, S_{\min(i,c)}\} \sim \mathrm{Exp}(\min(i,c)\mu).$$

  Therefore, the total departure rate from state i is $\min(i,c)\mu$, and hence

$$P_{i,i-1}(h) = \min(i,c)\mu\, h + o(h).$$

Hence, the process is a birth–death process with

$$\lambda_i = \lambda, \quad \mu_i = \min(i,c)\mu.$$

3.3 Stationary Distribution

Let

$$p_j = \lim_{t\to\infty} \mathbb{P}(X_t = j)$$

denote the stationary distribution.

Balance Equations

Using detailed balance,

$$\lambda p_{j-1} = \mu_j p_j,$$

where

$$\mu_j = \min(j,c)\mu.$$

Case 1: $0 \leq j < c$

Here, $\mu_j = j\mu$, so

$$\lambda p_{j-1} = j\mu\, p_j.$$

Thus,

$$p_j = \frac{\lambda}{j\mu} p_{j-1}.$$

Iterating,

$$p_j = \frac{1}{j!}\left(\frac{\lambda}{\mu}\right)^j p_0.$$

Case 2: $j \geq c$

Here, $\mu_j = c\mu$, so

$$\lambda p_{j-1} = c\mu\, p_j.$$

Thus,

$$p_j = \frac{\lambda}{c\mu} p_{j-1}.$$

Iterating from $j = c$,

$$p_j = \frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c \left(\frac{\lambda}{c\mu}\right)^{j-c} p_0.$$

Normalization

Using $\sum_{j=0}^{\infty} p_j = 1$, we obtain

$$p_0 \left[\sum_{j=0}^{c-1} \frac{1}{j!}\left(\frac{\lambda}{\mu}\right)^j + \frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c \sum_{j=c}^{\infty}\left(\frac{\lambda}{c\mu}\right)^{j-c}\right] = 1.$$

The second sum is geometric, giving

$$\sum_{j=c}^{\infty}\left(\frac{\lambda}{c\mu}\right)^{j-c} = \frac{1}{1-\frac{\lambda}{c\mu}}.$$

Thus,

$$p_0 = \left[\sum_{j=0}^{c-1} \frac{1}{j!}\left(\frac{\lambda}{\mu}\right)^j + \frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c \frac{1}{1-\frac{\lambda}{c\mu}}\right]^{-1}.$$

3.4 Final Expression

The stationary distribution is given by

$$p_j = \begin{cases} \dfrac{1}{j!}\left(\dfrac{\lambda}{\mu}\right)^j p_0, & 0 \le j < c, \\[2ex] \dfrac{1}{c!}\left(\dfrac{\lambda}{\mu}\right)^c \left(\dfrac{\lambda}{c\mu}\right)^{j-c} p_0, & j \ge c. \end{cases}$$

3.5 Stability Condition

For the series to converge, we require

$$\frac{\lambda}{c\mu} < 1.$$

This ensures that the system is stable.

3.6 Expected Number of Customers

Let $\rho = \frac{\lambda}{c\mu}$ denote the traffic intensity, and assume $\rho < 1$ for stability.

The expected number of customers in the system can be decomposed as

$$\mathbb{E}[N] = L = L_q + L_s,$$

where:

- L_q is the expected number of customers in the queue
- L_s is the expected number of customers in service

3.7 Expected Number of Customers: Derivation

Let $\rho = \frac{\lambda}{c\mu} < 1$.

Customers in Service

In steady state, the long-run departure rate must equal the arrival rate λ. Each customer in service is served at rate μ, so if L_s denotes the expected number of customers in service, then

$$\mu L_s = \lambda.$$

Thus,

$$L_s = \frac{\lambda}{\mu}.$$

Probability of Waiting

An arriving customer has to wait if and only if all c servers are busy, i.e., the system is in a state $j \geq c$.

Thus,

$$P_{\text{wait}} = \sum_{j=c}^{\infty} p_j.$$

Using the stationary distribution derived earlier, for $j \geq c$:

$$p_j = \frac{1}{c!} \left(\frac{\lambda}{\mu}\right)^c \left(\frac{\lambda}{c\mu}\right)^{j-c} p_0.$$

Hence,

$$P_{\text{wait}} = p_0 \cdot \frac{1}{c!} \left(\frac{\lambda}{\mu}\right)^c \sum_{j=c}^{\infty} \left(\frac{\lambda}{c\mu}\right)^{j-c}.$$

Letting $k = j - c$, we obtain

$$P_{\text{wait}} = p_0 \cdot \frac{1}{c!} \left(\frac{\lambda}{\mu}\right)^c \sum_{k=0}^{\infty} \left(\frac{\lambda}{c\mu}\right)^{k}.$$

This is a geometric series, giving

$$P_{\text{wait}} = p_0 \cdot \frac{1}{c!} \left(\frac{\lambda}{\mu}\right)^c \cdot \frac{1}{1 - \frac{\lambda}{c\mu}}.$$

Substituting the expression for p_0 from normalization yields the Erlang-C formula.

Expected Queue Length

The expected number of customers in the queue is

$$L_q = \sum_{j=c}^{\infty} (j - c) p_j.$$

Using the expression for p_j, we obtain

$$L_q = p_0 \cdot \frac{1}{c!} \left(\frac{\lambda}{\mu}\right)^c \sum_{j=c}^{\infty} (j - c) \left(\frac{\lambda}{c\mu}\right)^{j-c}.$$

Let $k = j - c$, giving

$$L_q = p_0 \cdot \frac{1}{c!} \left(\frac{\lambda}{\mu}\right)^c \sum_{k=0}^{\infty} k \left(\frac{\lambda}{c\mu}\right)^{k}.$$

Using the identity

$$\sum_{k=0}^{\infty} kx^k = \frac{x}{(1-x)^2}, \quad |x| < 1,$$

we obtain

$$L_q = p_0 \cdot \frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c \cdot \frac{\frac{\lambda}{c\mu}}{\left(1-\frac{\lambda}{c\mu}\right)^2}.$$

Recognizing that

$$P_{\text{wait}} = p_0 \cdot \frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c \cdot \frac{1}{1-\frac{\lambda}{c\mu}},$$

we simplify to

$$L_q = \frac{P_{\text{wait}} \cdot \frac{\lambda}{c\mu}}{1-\frac{\lambda}{c\mu}}.$$

Thus,

$$L_q = \frac{P_{\text{wait}} \cdot \rho}{1-\rho}.$$

Final Expression

Finally, the expected number of customers in the system is

$$\mathbb{E}[N] = L = L_q + L_s = \frac{\lambda}{\mu} + \frac{P_{\text{wait}} \cdot \rho}{1-\rho}.$$

3.8 Simulation Approach

To validate the analytical results derived for the $M/M/c$ queue, we simulate the system as a continuous-time Markov process.

At any state n:

- Arrivals occur at rate λ
- Departures occur at rate $\min(n, c)\mu$

Thus, the total transition rate is

$$\lambda + \min(n, c)\mu.$$

The simulation proceeds by:

- Sampling the time to the next event using an exponential distribution
- Determining whether the event is an arrival or a departure
- Updating the system state accordingly

3.9 Core Simulation

The system is simulated using the following structure:

```
arrival_rate <- lambda
service_rate <- min(n, c) * mu
total_rate <- arrival_rate + service_rate

dt <- rexp(1, total_rate)
```

The event type is determined probabilistically:

```
if(runif(1) < arrival_rate / total_rate){
  n <- n + 1
} else {
  n <- max(0, n - 1)
}
```

3.10 Estimation of Performance Measures

We estimate quantities using time averages. Let $\{t_i\}$ denote event times and $\{X_i\}$ the corresponding states.

Expected Number in System

$$L = \frac{\sum_i X_i \Delta t_i}{\sum_i \Delta t_i}$$

```
L_hat <- sum(states * durations) / total_time
```

Expected Queue Length

Only customers exceeding server capacity contribute to the queue:

$$L_q = \frac{\sum_i \max(X_i - c, 0)\Delta t_i}{\sum_i \Delta t_i}$$

```
queue <- pmax(states - c, 0)
Lq_hat <- sum(queue * durations) / total_time
```

Probability of Waiting

$$P_{\text{wait}} = \mathbb{P}(X_t \geq c)$$

```
P_wait_hat <- sum(durations[states >= c]) / total_time
```

3.11 Verification of Stationary Distribution

The stationary probabilities are estimated as:

$$\hat{p}_j = \frac{\text{time spent in state } j}{\text{total time}}.$$

```
p_hat <- sapply(0:max_state, function(j)
  sum(durations[states == j]) / total_time)
```

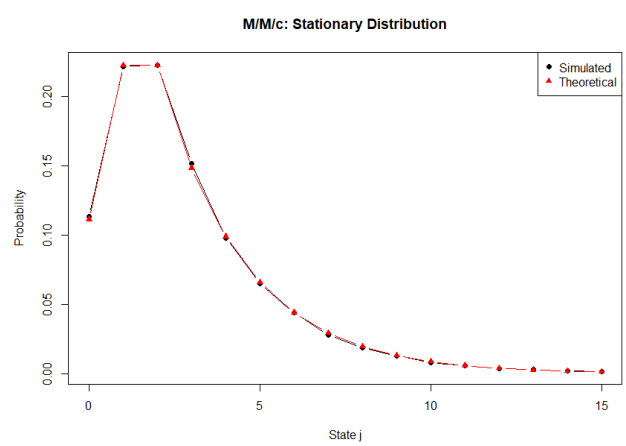
These are compared with the theoretical values derived earlier.

3.12 Results

For $\lambda = 4$, $\mu = 2$, and $c = 3$, we obtain the following:

Metric	Simulated	Theoretical
$L = \mathbb{E}[N]$	2.8733	2.8889
L_q	0.8778	0.8889
P_{wait}	0.4430	0.4444

A graphical comparison of $\hat{p}_j$ and p_j further confirms the match between simulation and theory.



3.13 Conclusion

The simulation validates the analytical expressions derived for the $M/M/c$ queue. The empirical estimates of all key performance measures closely match their theoretical counterparts.

This demonstrates convergence of the system to its stationary distribution and confirms the correctness of the balance-based derivations.

Furthermore, the results illustrate how increasing the number of servers reduces congestion, while higher arrival rates lead to significant increases in queue length and waiting probability.

Finally, this simulation framework provides a foundation for analyzing more complex systems such as $M/G/c$ queues, where closed-form solutions are not available.